

Total No. of Questions : 8]

SEAT No. :

PC-2489

[Total No. of Pages :2

[6354]-618R

B.E (Mechanical)

INDUSTRIAL ENGINEERING

(2019 Pattern) (Semester - VII) (402044 D) (Elective - III)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q5 or Q6, Q7 or Q8.
- 2) Figure to the right side indicates full marks.
- 3) Use of electronic is allowed.
- 4) Assume Suitable data if necessary.

Q1) a) Elaborate any four different types of material handling equipment's with suitable applications? [8]

b) Explain the term plant layout. Enlist types of plant layout. Describe the factors to be considered while finalizing the plant layout with suitable illustration. [9]

OR

Q2) a) Explain types of material flow patterns to achieve optimum flow of material in the plant? [8]

b) Compare line layout with function layout also explain objectives of facility layout planning [9]

Q3) a) Write a short note on Capacity Planning? Compare job Production and Mass Production. [9]

b) M/s XYZ Enterprises uses exponential smoothing with Smoothing Parameter 0.16. for estimating a demand following table shows the demand for 6 months the forecast for May is 545 units. Estimate the forecast for November. [9]

Month	Actual Demand
May	470
June	510
July	516
August	488
September	450
October	550

P.T.O.

OR

- Q4)** a) Write a short note on [9]
i) MPS
ii) MRP - I
b) Estimate the state forecast for the year 2013 using exponential smoothing forecaster. Take Smoothing Constant = 0.5 and the forecast for the year 2005 as 175×10^5 units. [9]

Year	2005	2006	2007	2008	2009	2010	2011	2012
Sale Rs.(x 10 ⁵)	195	170	160	175	190	195	200	195

- Q5)** a) Write a short note on ERP? Explain its benefits? Also explain different modules present in ERP [9]
b) Explain any three selective control techniques of inventory? [9]

OR

- Q6)** a) Write a short note on [9]
i) Supply chain strategies
ii) Economic order quantity.
b) The ABC enterprises use EOQ logic to determine the order quantity for its various components and are planning its orders. The Annual consumption is 95,000 units, Cost to place one order is Rs. 1400, Cost per unit is Rs. 60 and carrying cost is 6% of Unit Cost Find, [9]
i) EOQ,
ii) No. of order per year,
iii) Ordering Cost and Carrying Cost
iv) Total Cost of Inventory.

- Q7)** a) Define Job evaluation? What are its objectives? Explain procedure of job evaluation? [9]
b) Explain Rapid Entire body Assessment (REBA) with level of MSD risk. [8]

OR

- Q8)** a) Elaborate the term "merit rating"? Explain any 4 methods of merit rating? [9]
b) Write a short note on: [8]
i) KRA
ii) Principles of ergonomics

